Huawei Access Node WHI-ACC-50 Critical QoS Congestion Event SOP (ALARM_4054)

Node: WHI-ACC-50 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_4054

Description

The Huawei Access Node WHI-ACC-50 Critical QoS Congestion Event SOP (ALARM_4054) addresses a Critical QoS congestion event on the Huawei Access Node WHI-ACC-50 at the Whitefield East POP. This failure domain occurs due to the presence of microbursts, interface tail drops, and QoS queue starvation. The root cause is likely attributed to the inadequacy of the QoS configuration, which fails to account for the dynamic network traffic patterns, resulting in a build-up of packets in the QoS queues. The Huawei Access Node, being a key component in the access network, plays a critical role in ensuring the QoS requirements are met. However, the presence of microbursts and interface tail drops indicates a potential issue with the node's ability to handle bursty traffic, which may be exacerbated by inadequate buffer sizes or policing rates. Furthermore, the occurrence of QoS queue starvation implies that the node's QoS configuration is not effectively prioritizing critical traffic flows, leading to a degradation in network performance. The Huawei Access Node's reliance on the QoS configuration to manage traffic and ensure performance makes it susceptible to congestion events. To mitigate these issues, it is essential to thoroughly investigate the QoS configuration, policing rates, buffer sizes, and traffic patterns to identify potential bottlenecks and areas for improvement.

Pre-requisite Checks

- Verify that the redundant link status is operational and ready for use.
- Ensure that the remote hands are dispatched to the Whitefield East POP.
- Check the power supply and environmental conditions at the site.
- Verify that the Huawei Access Node WHI-ACC-50 is properly connected and configured.

Potential Root Causes

Cause	Probability
Inadequate QoS configuration	High
Policing rates too low	Medium
Buffer sizes too small	Low

Diagnostic Commands & Expected Outputs

```
Command: display queue statistics, Expected Output: Queue statistics
for all queues., Description: This command will provide detailed
statistics on the QoS queues, including the number of packets dropped,
the number of packets transmitted, and the average queue length. This
will help identify potential bottlenecks and areas for improvement in
```

```
the QoS configuration.

Command: display policing statistics, Expected Output: Policing statistics for all policing profiles., Description: This command will provide detailed statistics on the policing profiles, including the number of packets transmitted, the number of packets dropped, and the average policing rate. This will help identify potential bottlenecks and areas for improvement in the policing rates.

Command: display buffer statistics, Expected Output: Buffer statistics for all buffers., Description: This command will provide detailed statistics on the buffers, including the number of packets dropped due to buffer overflow, the number of packets transmitted, and the average buffer size. This will help identify potential bottlenecks and areas for improvement in the buffer sizes.
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Resolution Steps

Step 1: Step: Step 1: Verify QoS configuration, Command: display qos configuration, Description: Verify that the QoS configuration is adequate and properly prioritizes critical traffic flows. Check for any misconfigured or missing QoS policies, and adjust as necessary to ensure that the QoS requirements are met.

Step 2: Step: Step 2: Adjust policing rates, Command: configure policing profile, Description: Adjust the policing rates to ensure that the node can handle bursty traffic effectively. Check the policing statistics to ensure that the policing rates are adequate and not causing excessive packet drops.

Step 3: Step: Step 3: Increase buffer sizes, Command: configure buffer size, Description: Increase the buffer sizes to ensure that the node can handle bursty traffic without experiencing buffer overflows. Check the buffer statistics to ensure that the buffer sizes are adequate and not causing excessive packet drops.

Post-Check Verification

```
Command: display queue statistics, Expected Output: Queue statistics for all queues., Description: Verify that the QoS queues are operating within normal parameters and that there are no packet drops or excessive queue lengths.

Command: display policing statistics, Expected Output: Policing statistics for all policing profiles., Description: Verify that the policing profiles are operating within normal parameters and that there are no excessive packet drops or policing rates.

Command: display buffer statistics, Expected Output: Buffer statistics for all buffers., Description: Verify that the buffers are operating within normal parameters and that there are no excessive packet drops or buffer overflows.
```

Roll-back Procedures

- Step: Step 1: Restore original QoS configuration, Command: undo qos configuration, Description: Restore the original QoS configuration in case the new configuration causes issues with packet drops or excessive queue lengths.

- Step: Step 2: Restore original policing rates, Command: undo policing profile, Description: Restore the original policing rates in case the new policing rates cause issues with excessive packet drops or policing rates.

- Step: Step 3: Restore original buffer sizes, Command: undo buffer size, Description: Restore the original buffer sizes in case the new buffer sizes cause issues with excessive packet drops or buffer overflows.